AI CHATBOT FOR COLLEGE — ALL REVISED DIAGRAMS (6 PAGES)

Interaction boundaries and broad secondary database cylinder actor

1. Use Case Diagram (Broad Database Cylinder)

✔ Teacher Correction Applied: Broad database cylinder with zero stickman icon.

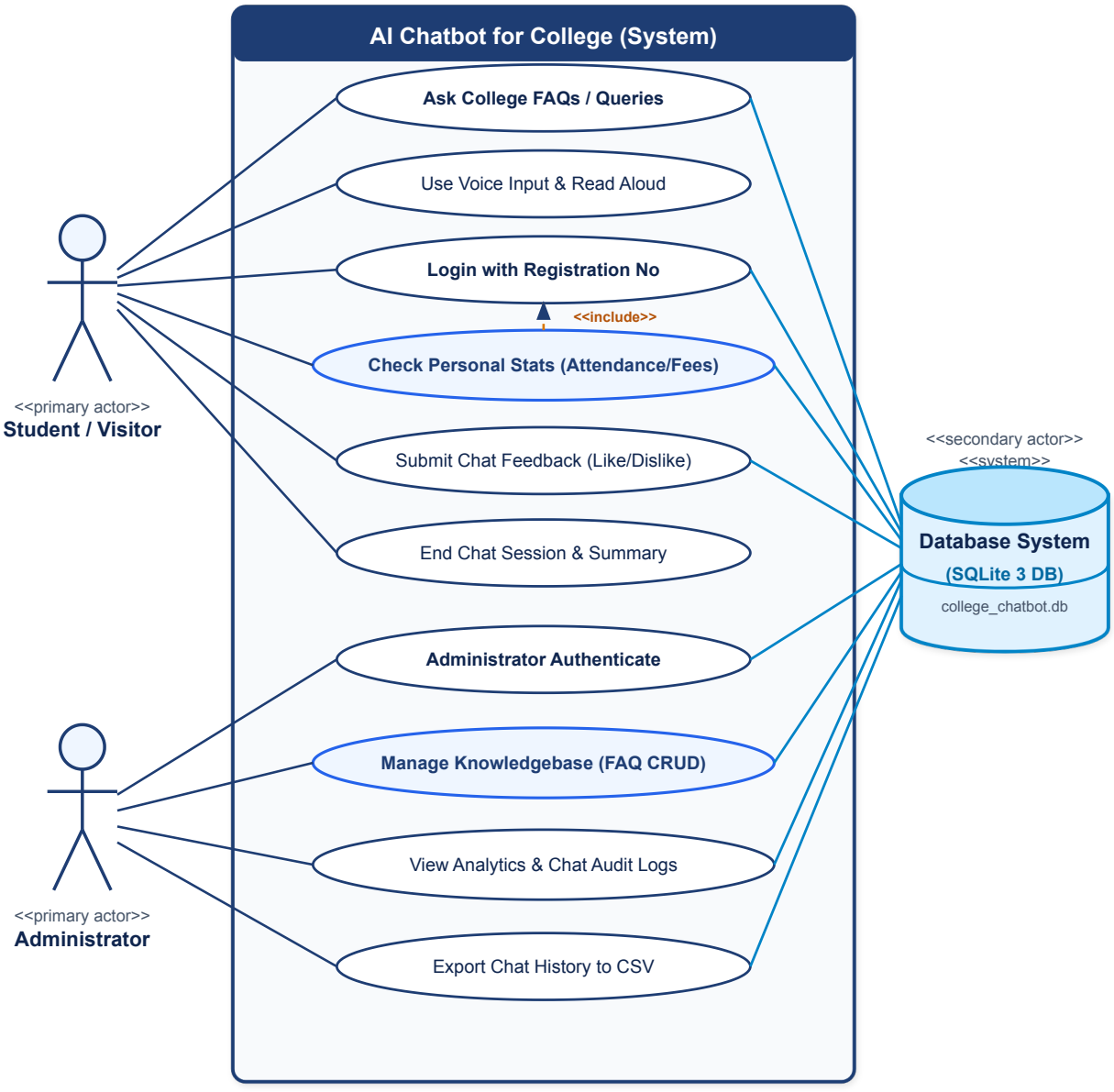


Fig 1: Use Case Diagram — primary actors and broad secondary database cylinder.

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Structural classes, attributes, methods, inheritance and associations

2. Class Diagram (Generalization & Associations)

✔ Teacher Correction Applied: Base User, Student/Admin inheritance, ChatHistory & CollegeInfo

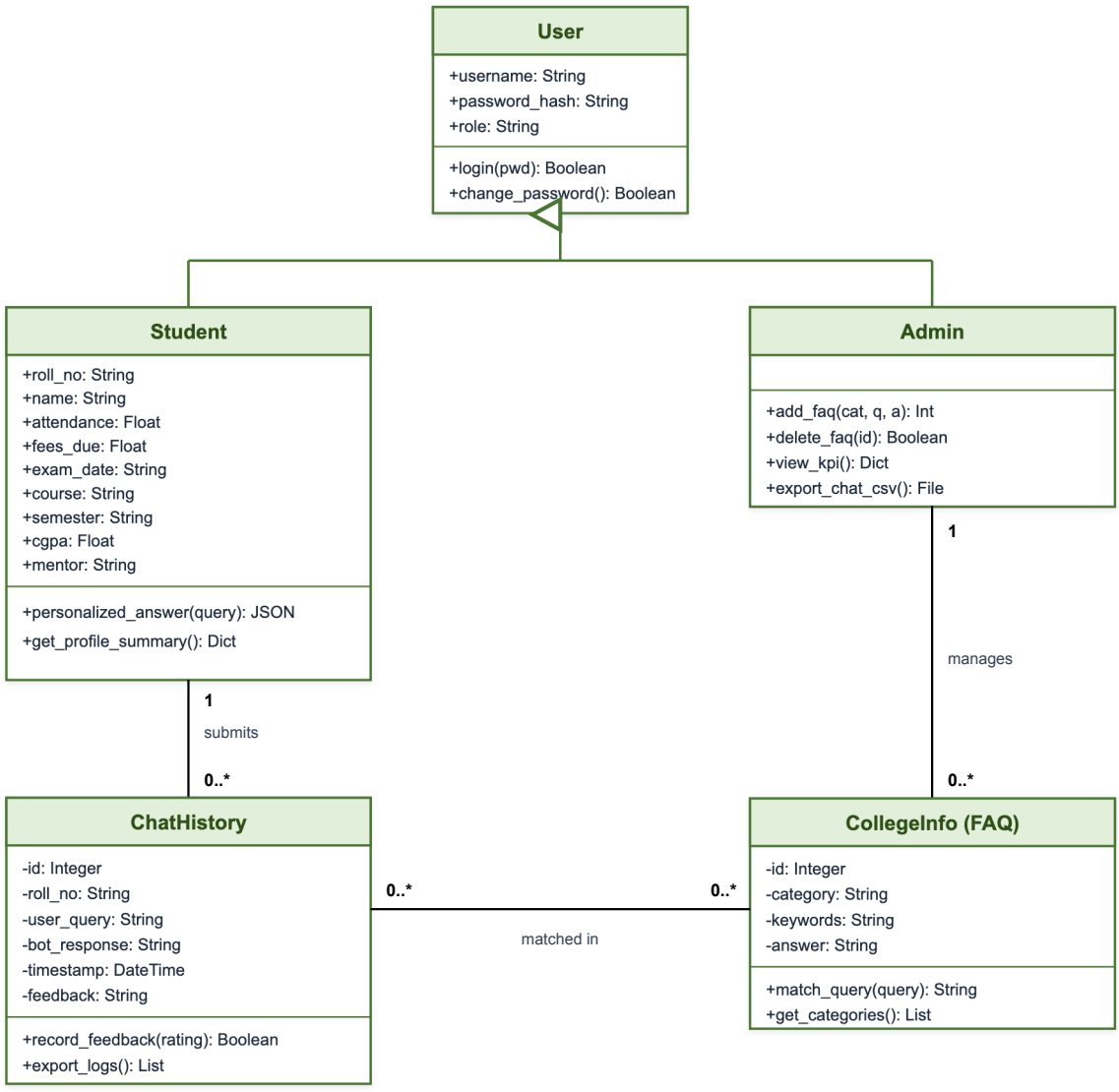


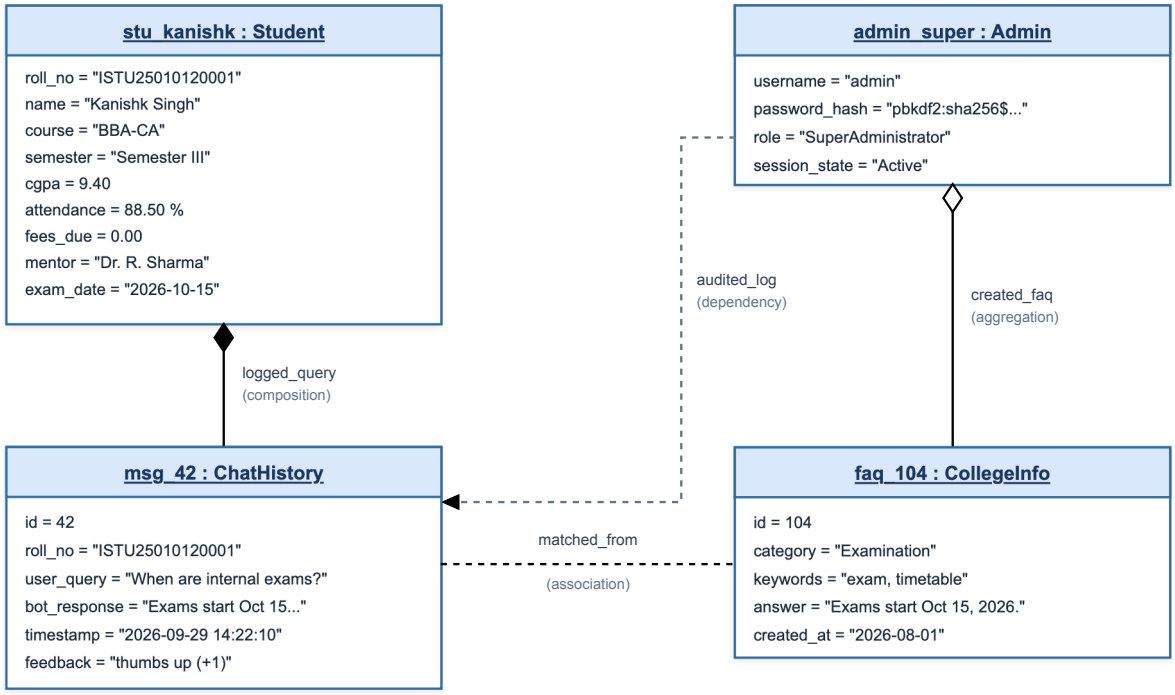
Figure 2: UML Class Diagram with Generalization, Associations, and Multiplicities

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Runtime execution snapshot with concrete object instances and slot values

3. Object Diagram (Concrete Runtime Instances & Relations)

✔ Teacher Correction Applied: Concrete instances with slot values, composition, aggregation & dependency



Object diagram notations and their uses

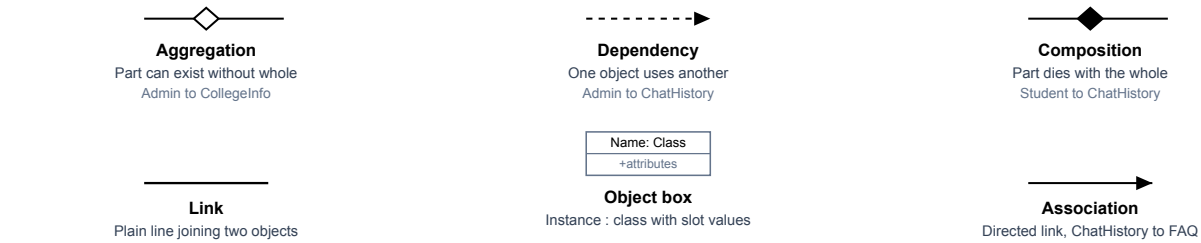


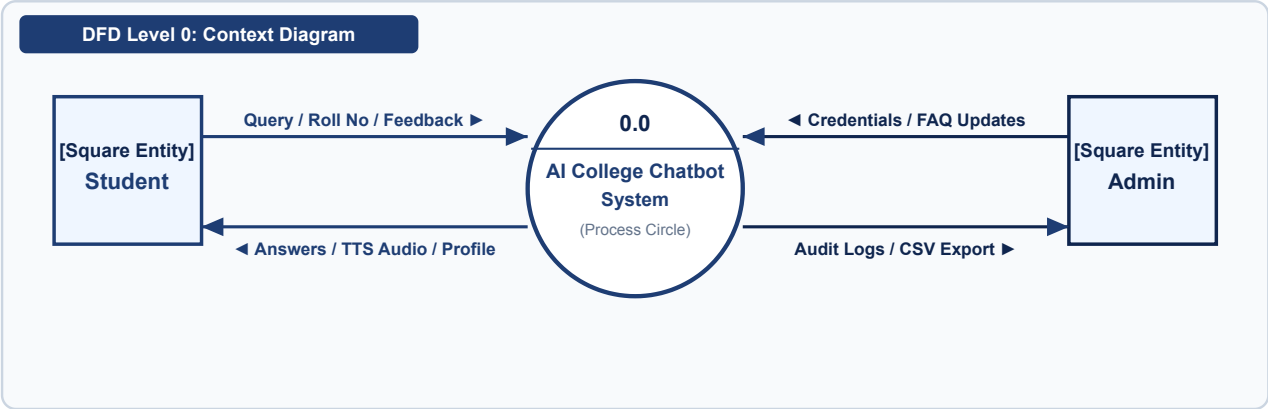
Figure 3: UML Object Diagram with Concrete Instances, Links, and UML Notations Legend

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Information flow with square entities and circle processes

4. Data Flow Diagram — DFD (Square Entities & Circles)

✔ Teacher Correction Applied: Square external entities & circle process symbols with solid flow arrows.



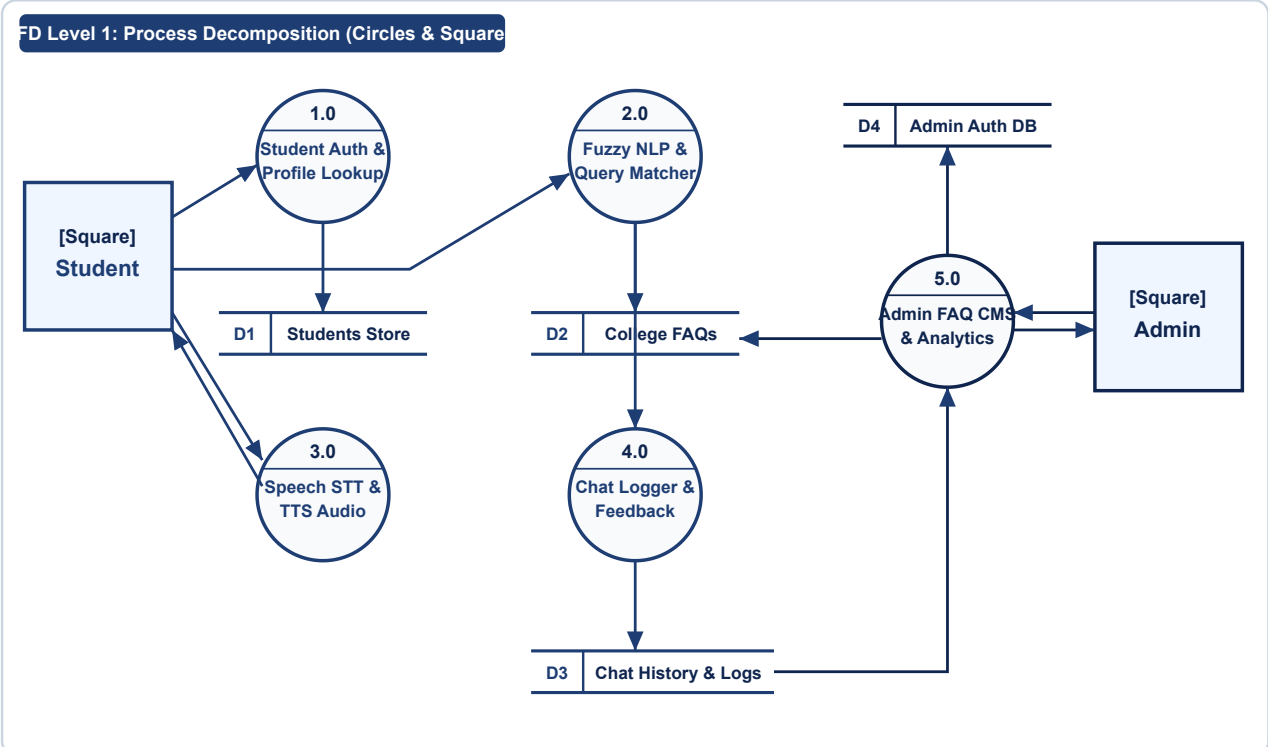


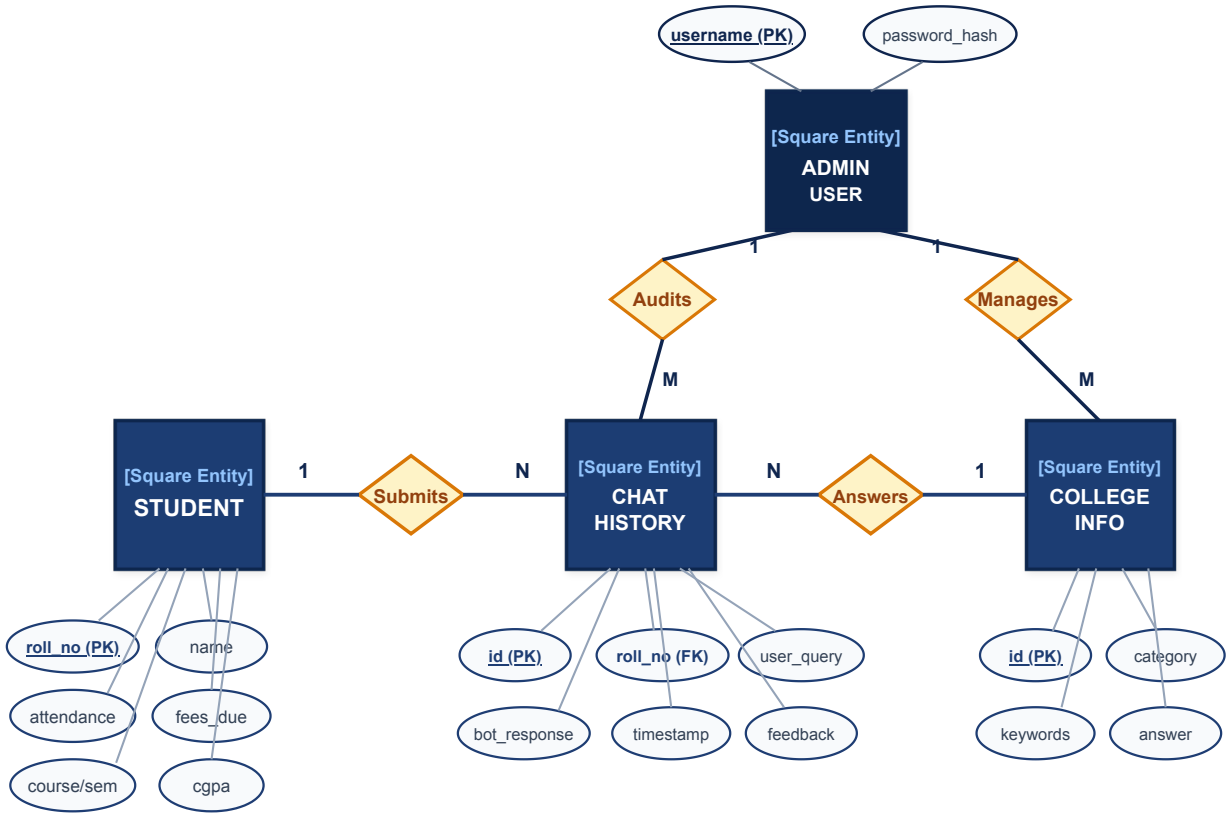
Fig 4: Data Flow Diagram (DFD) — square external entities and circle processes.

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Relational data schema with square entities and cardinalities

5. Entity-Relationship Diagram — ERD (Square Entities)

✔ Teacher Correction Applied: Square entity symbols for all 4 tables with clean non-crossing flow.



✔ **Standard Entity-Relationship Diagram Flow Summary:**

- Exact Square Entity Symbols: All entities (STUDENT, ADMIN_USER, CHAT_HISTORY, COLLEGE_INFO) rendered as square boxes.
- Left-to-Right Primary Data Flow: STUDENT [1] submits [N] CHAT_HISTORY, which answers [N:1] from COLLEGE_INFO.
- Top-to-Bottom Admin Control Flow: ADMIN_USER [1] audits [M] CHAT_HISTORY and manages [M] COLLEGE_INFO FAQs.
- Relational Integrity: Primary Keys underlined (<u>roll_no</u>, <u>id</u>, <u>username</u>); Foreign Key marked (roll_no FK).

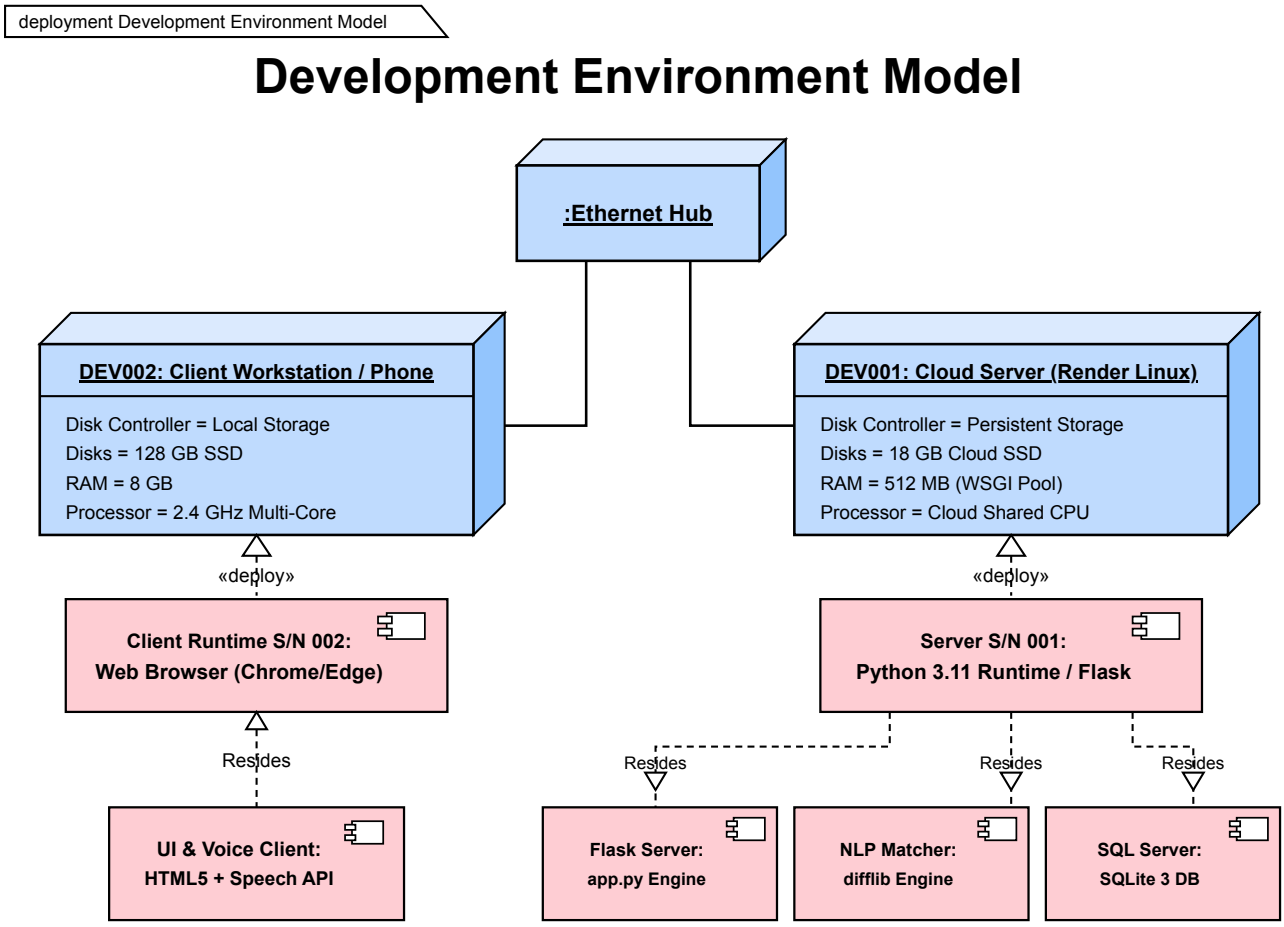
Fig 5: Entity-Relationship Diagram (ERD) — square entities and clean relational flow.

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Development Environment Model matching reference architecture

6. Deployment Diagram (Development Environment Model)

✔ Teacher Correction Applied: 3D Nodes, :Ethernet Hub, «deploy» arrows & component boxes.



UML Deployment Architecture Legend & Mapping:

- :Ethernet Hub: Central network routing gateway connecting client and cloud server hardware nodes.
- 3D Nodes (DEV001 & DEV002): Physical/virtual hardware devices with CPU, RAM, and Disk controller specifications.
- «deploy» & Resides: Formal deployment relationships indicating software execution runtime and resident artifacts.

Fig 6: Deployment Diagram — Development Environment Model matching reference architecture.